Yash Mehta

Machine Learning Research Engineer | AI Researcher | Deep Learning Developer

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RESEARCH PROFILE

Research-oriented Computer Science professional specializing in machine learning, deep learning, predictive modeling, and applied artificial intelligence. Experienced in designing experimental ML pipelines, developing hybrid neural network architectures, evaluating model performance, and implementing research prototypes. Co-authored publications in Springer Nature and IEEE conference proceedings, with research presented at SocProS-25, IIT Roorkee. Interested in applied AI research, experimental model development, and research engineering.

RESEARCH & TECHNICAL SKILLS

- *Programming*: Python, Java, C++
- *Machine Learning*: Scikit-learn, Random Forest, Classification, Feature Engineering, Model Evaluation
- *Deep Learning*: TensorFlow, CNN, LSTM, GRU, Vision Transformers (ViT)
- *Time-Series Research***:** ARIMA, Prophet, LSTM, GRU, Transformers
- *NLP*: TF-IDF, Sentiment Analysis, Text Classification
- *Computer Vision*: OpenCV, Image Processing, FRIDA2 Dataset
- *Data Analysis*: Pandas, NumPy, Statistical Evaluation
- *Research Tools*: Jupyter Notebook, Google Colab, Git
- *Application Development*: Flask, REST APIs

RESEARCH EXPERIENCE

Fake News Detection Using Machine Learning and Sentiment Analysis | IIT Roorkee | 2025

- *Designed and implemented a machine learning pipeline for fake news classification using TF-IDF and Random Forest.*
- *Integrated sentiment analysis to enrich textual feature representation.*
- *Developed a Flask web application to demonstrate the research implementation.*
- *Achieved 87% validation accuracy.*
- *Presented the research at SocProS-25, IIT Roorkee.*
- *Subsequently contributed to the Springer publication in Real-World Applications of Machine Learning.*

Fog Detection Using Hybrid Deep Learning Models | IEEE ICCCA | 2025

- *Investigated hybrid deep learning approaches for fog detection using a multi-modal CNN-LSTM research framework.*

- *Worked with the FRIDA2 dataset for visual fog detection experiments.*
- *Evaluated CNN and Vision Transformer-based approaches.*
- *Reported a 15% improvement in accuracy on real-world haze datasets.*
- *Integrated OpenCV for image processing and inference.*

Blockchain in Autonomous Vehicles | Bennett University | 2024

- *Investigated blockchain-based security mechanisms for autonomous vehicle communication.*
- *Simulated protocols and evaluated spoofing-related risks in test environments.*
- *Reported a 35% reduction in spoofing risks under experimental conditions.*

SELECTED RESEARCH PROJECTS

Amazon Stock Price Prediction | 2025

- Developed comparative forecasting models using ARIMA, LSTM, GRU, and Transformer architectures.
- Achieved RMSE of 0.03285 in forecasting experiments.
- Implemented a custom multivariate TimeSeriesTransformer.

Depression Detection via Facial Cues | 2024

- Developed a CNN-based facial image classification prototype using OpenCV.
- Achieved 82% classification accuracy in project evaluation.
- Implemented a Flask-based demonstration interface.

RESEARCH PUBLICATIONS

1) Fake News Detection Using Machine Learning and Sentiment Analysis Integrated in a Flask Web Application.

In: Pant, M., Deep, K., Nagar, A.K. (eds.), Real-World Applications of Machine Learning. Springer, Singapore, 2026.

DOI: https://doi.org/10.1007/978-981-95-7134-5_23

Earlier Conference Contribution: SocProS-25, Soft Computing for Problem Solving, IIT Roorkee, 2025. Book of Abstracts, page 90.

2) Fog Detection Using Hybrid Deep Learning Models: A Multi-Modal CNN-LSTM Approach.

Mehta et al., IEEE 7th International Conference on Computing, Communication and Automation (ICCCA), 2025.

DOI:10.1109/ICCCA66364.2025.11325690

IEEE Xplore: https://ieeexplore.ieee.org/document/11325690

TECHNICAL PROJECTS

Retail Analytics Dashboard - Vishleshan

- Developed a Flask and MySQL-based analytical dashboard using Chart.js.
- Demonstrated the project to 200+ industry experts at Bennett University.

FIR Legal Assistant - LexHack

- Developed an AI-powered legal assistance prototype using Flask, Gemini API, and Google Maps API.
- Received the Innovation & Impact award.

INTERNSHIP EXPERIENCE

Software Intern | MNNIT Allahabad | 2024

- Implemented Flask REST APIs for academic services.
- Improved backend throughput by 20%.
- Participated in Agile development activities.

CERTIFICATIONS

- NVIDIA - Fundamentals of Deep Learning
- IBM - IBM Machine Learning Professional Certificate
- Google Cloud - Introduction to Generative AI
- IBM - Generative AI Fundamentals
- IBM - AI Foundations for Business
- Google - Google Data Analytics Professional Certificate
- IIT Delhi - Quantum Computing & QKD

EDUCATION

B.Tech. - Computer Science & Engineering

Bennett University | 2022–2026 | CGPA: 8.26

Class XII - CBSE | 2021 | 92.4%

Class X - CBSE | 2019 | 85.8%

Maharshi Patanjali Vidya Mandir